3 Processing Model

3.1 Model (Broadly Speaking)

Model

Abstract representation of an existing entity or an entity to be created.

3.2 Type Of Model

3.2.1 Descriptive & Prescriptive Model

Descriptive & Prespective Model

- **Descriptive Model** : if a model represents an **existing entity** , e.g : biology describes existing creatures.
- **Prescriptive Model** : if it represents an **entity to be created** , e.g : blueprint of building prescribes an entity to be created.

3.2.2 Difference Between Descriptive & Prescriptive Model

Difference

The difference resides solely in the **purpose but not necessarily in the content** of the model.

3.3 Software Process Model

Software Process Model

- **Describes The Way Software Is To Be Developed** : in this case the software doesn't exist already therefore it's a **prescriptive model** to design the entity to be created.
- **Describes How The Software Is Actually Developed** : the software already exists so it's a **descriptive model**, its purpose is to **analyse & improve this process**.

Note

Software processes have the property of utilizing both descriptive & prescriptive models.

3.4 Stachowiak's General Model Theory (GMT)

GMT

In this theory the model is characterized by 3 properties :

- **Mapping Property** : a model is **mapping or representation** of some entity.
- **Reduction Property** : a model is a **reduction of the represented entity** and **doesn't contain all attributes** of the original (**an abstraction**).
- **Pragmatic Property** : a model is created for a **certain purpose** which implies that **the perfect model doesn't exist**, moreover one entity **may have different models** each contributing to a different purpose e.g building blueprints : architectural (layout/design), structural (foundation/framing), MEP (mechanical, electrical, plumbing).

3.5 Meta Model

Meta Model

In order to describe a model we need to define **how is to be done** and what terminologies may be used, in other words we need a model for the model (a meta model).

3.6 ISO Meta Model Definition

3.6.1 ISO 24744

ISO 24744

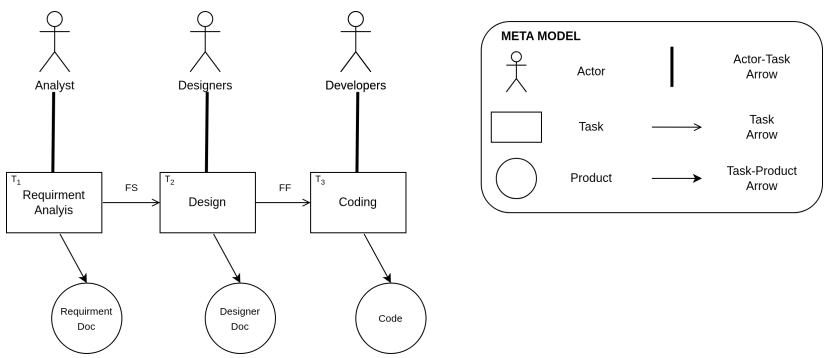
Meta model is specification of the concept relationships on the rules that are used to define a mythodology.

3.6.2 ISO 15909

ISO 15909

Meta model is a model defining the concept and the relation for some modeling notation.

3.7 Simple Example Of Meta Model



3.8 Some Terminology

3.8.1 Performed Process V.S Process Model

Performed Process V.S Process Model

Very common mistake is to refer to the process when actually talking about the process model, the term process refers to the process actually performed which may or may not be the same as described in the process model a process may be modeled but not yet performed or it may be performed but not modeled

3.8.2 Process As In V.S Process To Be

Process As In V.S Process To Be

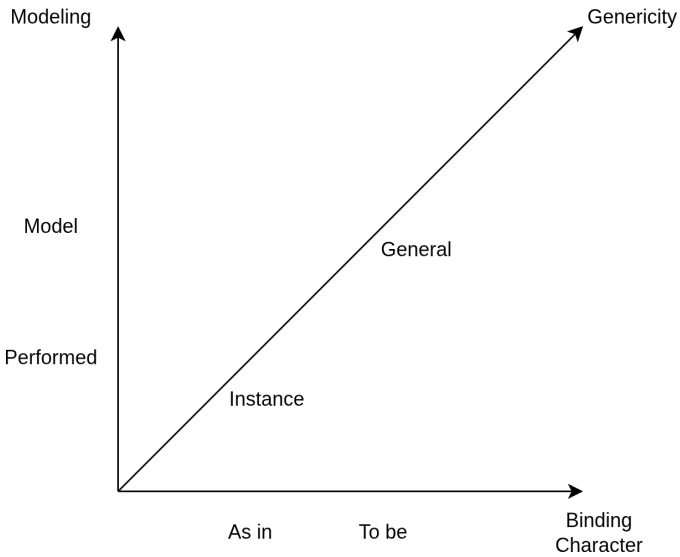
A process model may be descriptive describing the process as is or it may be prespective describing the process to be.

3.8.3 General Process V.S Process Instance

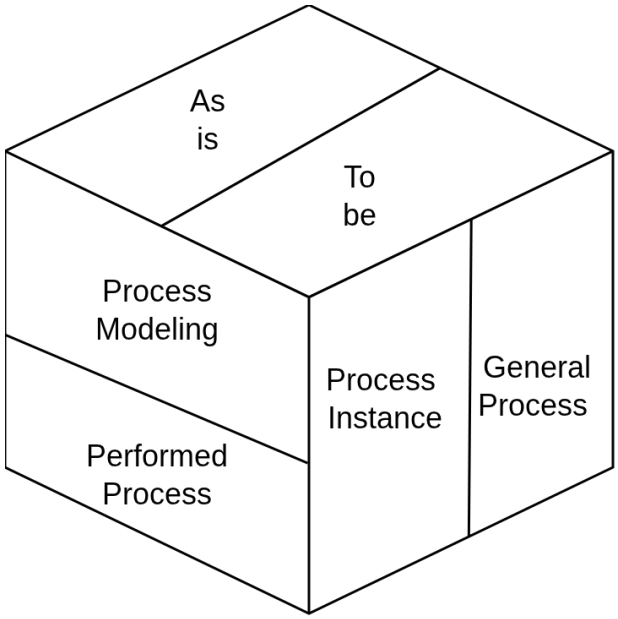
General Process V.S Process Instance

Processes exist on 2 different levels :

- Level of the individual instance where the process is performed once.
- Level of general process where the process is performed repeadetly across different instances.



Process Graph



Process Cube

3.9 Standard Processes & Tailoring

3.9.1 Standard Process

Standard Process

ISO 33001 : set of definitions of the processes used to guide processes in organization

The standard process is often defined on the level of an organization or soome time more general as public standard norm and applies to all instances of process in all or at least many projet of an organization

Its location in the process cube is : General process, To be, Process Model

3.9.2 Process Tailoring

Process Tailoring

Making , altering or adapting a process description for a particular end

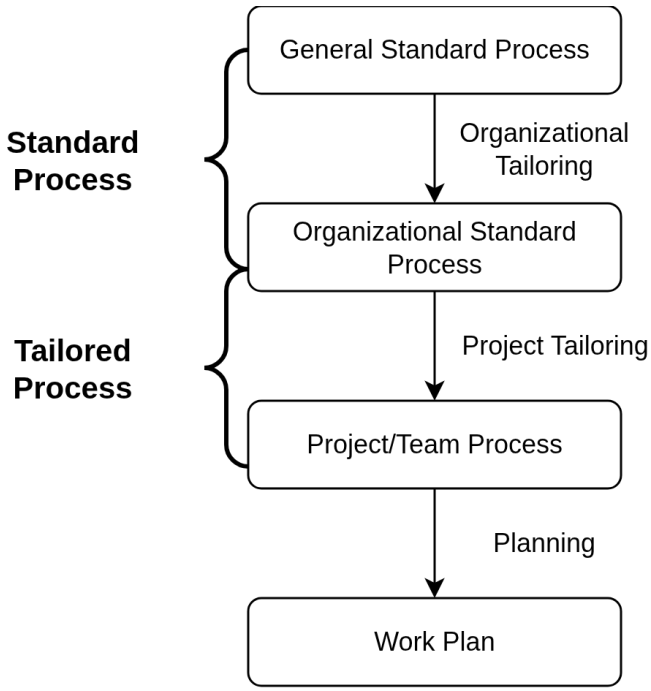
3.9.3 Tailored Process

Tailored Process

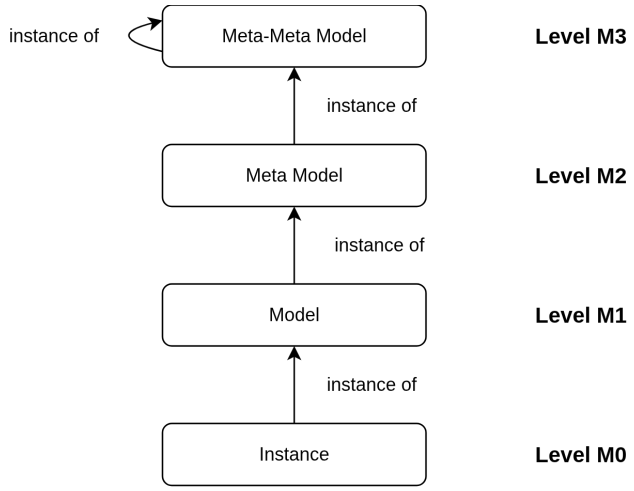
ISO 33001 : process developed by tailoring a standard process, tailoring may occur on different levels :

- From General standard process to organizational standard process
- From Organizational standard process to Project/Team process

Its location in the process cube is : General process, To be, Process Model for 1st level (organizational standard process) and Instance process, As is, performed process for 2nd level (Project/Team)



3.10 The Fourth Levels Of Modelisation

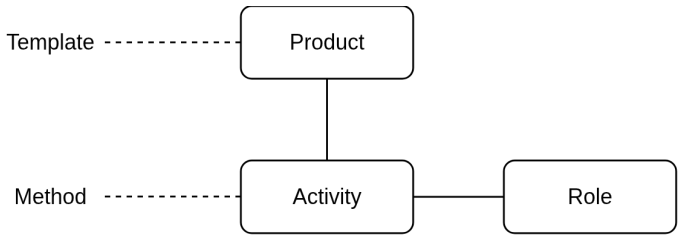


3.11 Core Contents Of Software Process Model

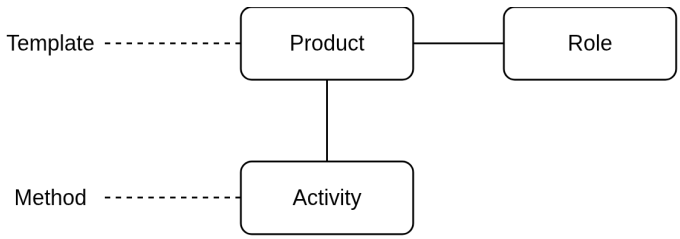
Core Contents

The core elements of major software process model are : **the manipulated product**, **the activities** manipulating these product and **roles** responsible for the work.

3.12 Actvity Oriented Representation



3.13 Product Oriented Representation



3.14 Further Contents Of Software Process Model

Further Contents

- Description of an activity or a group of activities
- Description of the product flow
- Descri[tion of control flow
- Description of the refinement

3.15 Kinds Of Software Process Modeling

Kinds

- **From Scratch** : Based on nothing built on your own
- **Process Model Component** : Like electronic we take already existing component without changing & just assembling
- **Process Patterns** : Like design pattern we have to adapt **Process Framework** that are nearly finished solution that we inject our parameters.

3.16 2 Different Complementary Ingeneering

2 Ingeneering

- **Modeling For Reuse** : Models reusable element that are generic enough not to small (it would be as builing from scratch) and not too big (ease of flexibility)
- **Modeling By Reuse** : use already existing element to help on modeling but we need to know that they exist, and how to properly use them and how to adapt them according to our context

3.17 Process Model Execution & Orchestration

Execution & Orchestration

Execution is all the activities and workflows carried out by the responsible actors. Orchestration coordinates who does what and when, ensuring the process runs as defined.

3.18 Concept Of Collaboration

Collaboration

Since multiple actors work on shared artifacts, collaboration tools are needed to coordinate their work. for example Git, which lets developers work concurrently without conflicts.